

Section 3.1 The Power Rule

Ex ① Use the definition of the derivative to find the derivative of $f(x) = x^3$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(3x^2 + 3xh + h^2)}{h}$$

$$= \lim_{h \rightarrow 0} 3x^2 + 3xh + h^2$$

$$= 3x^2$$

So, if $f(x) = x^3$ then

$$f'(x) = 3x^2$$

Power Rule

If $f(x) = x^n$ ← n is any number

$$\text{then } f'(x) = n \cdot x^{n-1}$$

Ex ② $f(x) = x^7 + 5x^4 + \sqrt{x} + \frac{1}{x}$

Rewrite all terms as $x^n \dots$

$$f(x) = x^7 + 5x^4 + x^{1/2} + x^{-1}$$

Now use the Power Rule on each term.

$$f'(x) = 7x^6 + 20x^3 + \frac{1}{2}x^{-\frac{1}{2}} - 1 \cdot x^{-2}$$

Don't leave negative exponents in the answer.

$$f'(x) = 7x^6 + 20x^3 + \frac{1}{2\sqrt{x}} - \frac{1}{x^2}$$

Ex ③ Given $y = 4x - 7$, find y'

$$y = 4x^1 - 7x^0 \quad \text{is another way to write the function}$$

$$\text{Then } y' = 1 \cdot 4x^0 - 0 \cdot 7x^{-1}$$

$$y' = 4$$

This says the slope of $y = 4x - 7$ is 4.

We can summarize this as follows:

FACTS If a is any constant

$$\frac{d}{dx}(a \cdot x) = a$$

$$\frac{d}{dx}(a) = 0$$

There are other shortcuts we can create using the power rule!

$$\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$$

$$\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$$

Proof If $y = \sqrt{x}$, then we can write that as $y = x^{\frac{1}{2}}$.

$$y' = \frac{1}{2} x^{-\frac{1}{2}}$$

$$y' = \frac{1}{2x^{\frac{1}{2}}} \text{ or } y' = \frac{1}{2\sqrt{x}}$$

Ex ④ when is $f'(x)$ undefined?

$$f(x) = (x-2)^{2/3} + 1$$

$$f'(x) = \frac{2}{3}(x-2)^{-1/3}$$

or $f'(x) = \frac{2}{3(x-2)^{1/3}}$

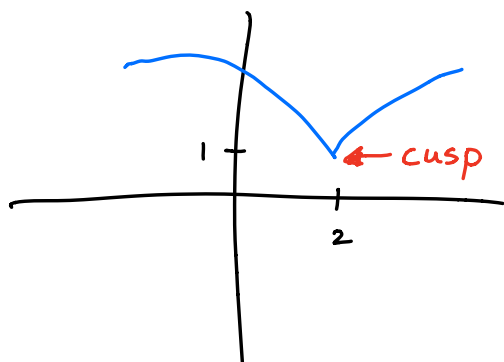
And this is undefined when

$$3(x-2)^{1/3} = 0$$

$$(x-2)^{1/3} = 0$$

$$x-2 = 0$$

$$\boxed{x = 2}$$



Ex ⑤ when is $f'(x) = 0$?

Given $f(x) = x^3 - 3x^2$

Use the Power Rule...

$$f'(x) = 3x^2 - 6x$$

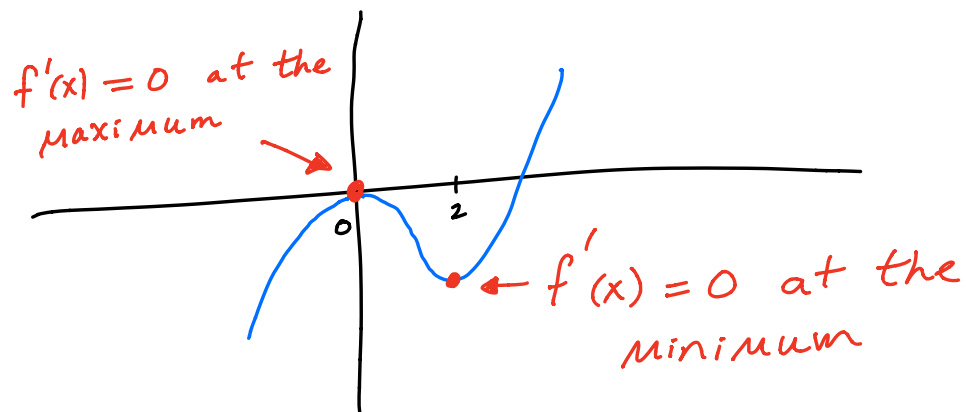
$$0 = 3x^2 - 6x$$

$$0 = 3x(x-2)$$

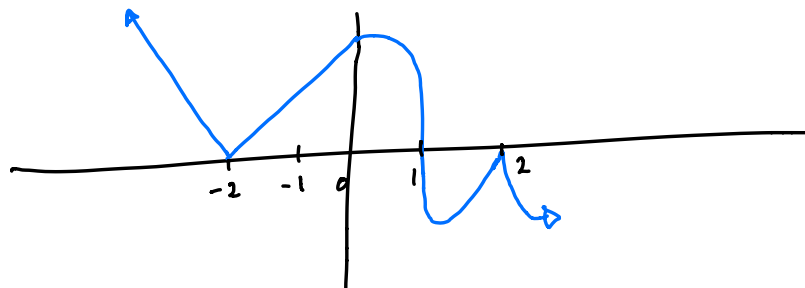
$$3x=0 \quad \text{or} \quad x-2=0$$

$$x=0$$

$$x=2$$



Ex ⑥ Where is $f'(x)$ undefined?



$f'(x)$ does not exist at

$$x = -2, x = 0, x = 1, x = 2$$

What about e^x ?

$$\frac{d}{dx}(e^x) = e^x$$

Ex ⑦ $y = 7 + 15e^x$

$$y' = 15e^x$$

When does $y' = 0$?

$$0 = 15e^x$$

$0 = e^x$ has no solution

